Lispec-44 Premix Powder

Lispec-44 Premix Powder appears to be an almost white color powder.

Composition:

Each kg contains

lincomycin (as lincomycin hydrochloride)	22 g
spectinomycin (as spectinomycin hydrochloride)	22 g

Pharmacodynamics:

Lincomycin belongs to the group of lincosamides and is produced by *Streptomyces lincolnensis*. The antibacterial activity of Lincomycin is primarily bacteriostatic and is active against gram-positive anaerobes and aerobes, gram-negative anaerobes and *Mycoplasma*. Lincomycin is ineffective against most Gram-negative pathogens (such as *Enterobacteriaceae*). Development of resistance against lincomycin, especially *staphylococci*, is observed. Lincomycin inhibits bacterial protein synthesis through reversible binding to the 50S subunit of bacterial ribosomes. Depending on sensitivity and concentration, the antibacterial effect is thus bacteriostatic or bactericidal.

Spectinomycin is an aminocyclitol belonging to the group of aminoglycoside antibiotics and is produced by *Streptomyces spectabilis*. The antibacterial activity of spectinomycin targets a range of Gram-positive and Gram-negative bacteria as well as *Mycoplasma*. Spectinomycin acts bacteriostatically by inhibiting bacterial protein synthesis at the 30S subunit of bacterial ribosomes. Lincomycin and spectinomycin have been shown to be synergistic in vitro against anaerobes implicated in the pathogenesis of swine dysentery. Spectinomycin is expected to have a high resistance rate. Under therapy, resistance develops rapidly according to the "one step type". Existing resistance to spectinomycin may also extend to aminoglycosides, erythromycin and tylosin.

Pharmacokinetics:

In pig, approximately 53% of lincomycin is bioavailable after oral administration. The elimination of lincomycin and metabolites occurs mainly via the liver and via milk in the case of lactating animals. Spectinomycin is absorbed after oral administration to about 10%. More than 90% remain in the intestinal tract and are excreted in the faeces.

Indications:

For the treatment of swine dysentery, caused by Brachyspira hyodysenteriae, which is complicated by spectinomycin-sensitive pathogens of the dysentery-associated flora (commensals, such as E. coli and Campylobacter spp).

Recommended Dosage:

For administration via feed.

For the treatment of individual animals (pigs).

4.4mg lincomycin/ kg body weight/ day and

4.4mg spectinomycin/ kg body weight/day equivalent to

200mg LISPEC-44 Premix Powder/ kg body weight/ day

The treatment duration is 21 days.

LISPEC-44 Premix Powder should be freshly added to a suitable quantity of feed before incorporation into the final mix so that complete mixing is achieved. It is important to ensure that the intended dose is taken completely.

In animals with a clearly disturbed general condition and/ or inappetence, preference must be given to a parenteral preparation.

If no significant improvement of the disease has occurred after a maximum of three days of treatment, the diagnosis must be checked; if necessary, the therapy should be changed.

Due to the presence of very high resistance rates to lincomycin and spectinomycin, treatment should only be given after evidence of the susceptibility of *Brachyspira hyodysenteriae* and the laboratory-confirmed pathogens of the accompanying dysentery flora (commensals such as *E. coli* or *Campylobacter spp*.).

Special precautions for users

People with known hypersensitivity to the active substances should avoid contact with the veterinary medicinal product.

In order to prevent sensitization or contact dermatitis, direct contact with the skin and mucous membranes and inhalation during working or processing and / or use should be avoided. Wear protective clothing such as safety glasses, dust mask and impermeable gloves.

In case of contact with skin or eyes, rinse thoroughly with water. If, after exposure, symptoms such as rash or persistent eye irritation occur, seek medical advice and show the package leaflet or label. Wash hands after use.

Interactions with Other Medicaments:

There is complete cross-resistance between lincosamides (lincomycin and clindamycin) and partial cross-resistance to macrolide antibiotics such as erythromycin, kitasamycin, spiramycin and tilmicosin. Partial cross-resistance may exist within the group of aminoglycoside antibiotics.

An antagonism was detected between lincomycin and erythromycin.

When co-administered with anesthetics or neuromuscular blocking agents (tubocurarine, gallamine, pancuronium), lincomycin potentiates the curare-like effects of these muscle relaxants.

Enteric absorption of lincomycin is reduced by about 50% with simultaneous food intake, and worsened by kaolin or pectin.

Mixing with other medicinal products should be avoided due to incompatibilities, eg in vitro incompatibilities of lincomycin with penicillins and kanamycin.

Statement on Usage During Pregnancy and Lactation:

Special care should be taken when using Lispec-44 Premix Powder in lactating animals as there may be potential gastrointestinal side effects of lincomycin in very young (suckling) animals.

Side effects, Overdose and Treatment:

Following lincomycin use, occasionally diarrhea, vomiting and anorexia, rarely redness and restlessness may occur.

In pigs, slight redness or swelling of the anal and vulvar areas may occur after oral administration within the first days of treatment. These symptoms are transient and resolve spontaneously within 5 to 8 days.

Allergic reaction.

Neuromuscular blockages, which are only partially removed by indirect-acting parasympathomimetics (eg neostigmine), as well as by calcium, can occur in isolated cases.

In isolated cases, agranulocytosis, leukopenia, thrombocytopenia, increase in AST activity in serum, influence on cardiac conduction velocity and hypotension have been observed.

If gastrointestinal disturbances occur shortly after the start of treatment or worsening of existing diarrhoea, treatment discontinuation or change is indicated.

In allergic reactions, the drug should be discontinued and it may be treated symptomatically: In case of anaphylactic shock: epinephrine (adrenaline) and iv glucocorticoids. For allergic skin reactions: glucocorticoids.

Withdrawal Period (s):

Pig:

Edible tissue: 2 days

Contraindication:
Do not use in case of hypersensitivity to lincomycin, spectinomycin and clindamycin or hypersensitivity to any of the excipients. Do not use in case of known resistance to the active substances. Do not use on horses, rabbits, hamsters, guinea pigs, chinchillas and ruminating animals (risk of severe colitis).
This product should not be used in case of liver dysfunction.
Concomitant use with macrolide antibiotics is not useful due to the identical point of action in bacterial metabolism.
Do not use concomitantly with anesthetics or neuromuscular blocking agents. Do not use on neonates because of possible toxic effects.

Warning and Precautions:
Special warnings for each target species
In rare cases respiratory arrest may occur in connection with anesthesia (barbiturates).
Special precautions for use in animals
The oral use of Lispec-44 Premix Powder is indicated only in pigs.
In other species, lincomycin can cause severe gastrointestinal disturbances.
In farms affected by swine dysentery, efforts should be made to avoid repeated use of the product by optimizing farm management, eg in animal husbandry and hygiene measures. A refurbishment is to be considered.

Storage Condition:
Store below 30°C. Protect from heat and sunlight exposure.
Keep out of reach of children/ Jauhkan dari kanak-kanak.
Any unused veterinary medicinal product or waste materials derived from such veterinary medicinal products should be disposed of in accordance with local requirements.
Shelf life after first opening: 7 days.

Packaging Available: 10 Kg

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Product Holder/ Manufacturer:
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